

Data Structures (SOC2010)

Submission Deadline: **15 /Nov/ 2023 (23:00 Hr)**

Problems based on Encoding and Threaded Binary Tree

NOTE: [Submissions **not conforming to the given format will not be evaluated**]

- You are required to **take print out of this sheet** and solve (**handwritten**) the questions on sheet.
- **Copying** of solution is strictly prohibited. In case it is found, then zero marks will be awarded.
- Late submissions are not accepted.

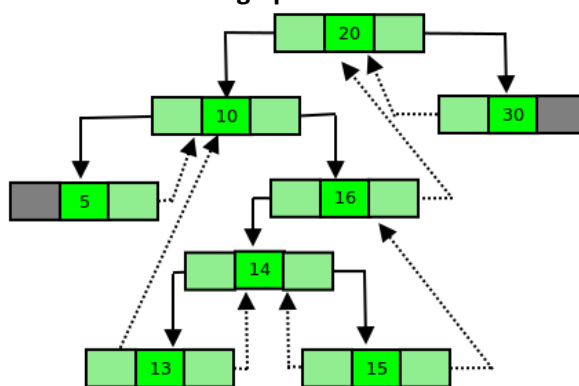
Questions

1. For the string **aaaapppssappss**, find the number of bits required if encoded using

- a. Default Encoding b. Uniform length encoding c. Variable length encoding

Answer

2. Perform the following operations on the following Threaded Binary tree



Delete 30

Note

- All the operations are performed on the given tree (not on the tree obtained after the operation)
- Show graphically the tree obtained after the operation (use five fields for every node)
- Just write necessary statements required to perform the given operation

Delete 10